

MATH10073: Linear Programming, Modelling and Solution (LPMS)

Notes and examples

Julian Hall

1 Slack variables

Slack variables are introduced to transform the “general” LP studied in this course

$$\text{maximize } f = \mathbf{c}^T \mathbf{x} \quad \text{subject to } A\mathbf{x} \leq \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0}$$

into **standard form**

$$\begin{array}{ll} \text{maximize} & f = \bar{\mathbf{c}}^T \mathbf{x} \\ \text{subject to} & \bar{A}\mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{array}$$

where $\bar{A} = [A \ I]$ and $\bar{\mathbf{c}} = \begin{bmatrix} \mathbf{c} \\ \mathbf{0} \end{bmatrix}$. This is done so that systems of general inequalities and bounds (which are hard to analyse) are transformed into systems of equations (which are well studied) and bounds on their variables.

You might ask the following. “If an LP is modelled using equations, why convert them into a pair of inequalities (as discussed in Workshop 1), and then add two slack variables to convert them back into equations?” The answer is that, in practice, LP solvers handle the most general form of LP

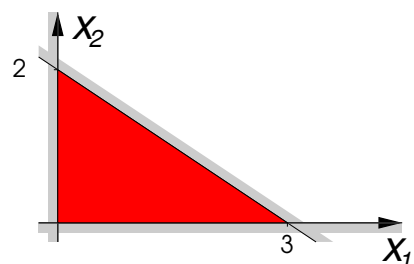
$$\text{maximize } f = \mathbf{c}^T \mathbf{x} \quad \text{subject to } \mathbf{L} \leq A\mathbf{x} \leq \mathbf{U}, \quad \mathbf{l} \leq \mathbf{x} \leq \mathbf{u}$$

where components of $\mathbf{L}$, $\mathbf{U}$, $\mathbf{l}$, and $\mathbf{u}$ may be infinite, and components of $\mathbf{L}$ and $\mathbf{U}$ may be equal.

Example 3

For the LP

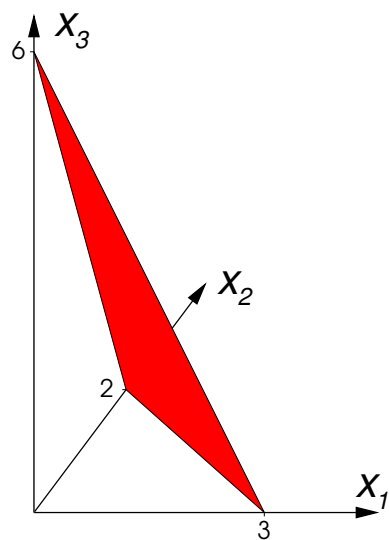
$$\begin{array}{ll} \max & f = 3x_1 + 5x_2 \\ \text{s. t.} & 2x_1 + 3x_2 \leq 6 \\ & x_1 \geq 0 \quad \text{and} \quad x_2 \geq 0 \end{array}$$



Add slack variable x_3 to give

$$\begin{array}{ll}\max & f = 3x_1 + 5x_2 \\ \text{s. t.} & 2x_1 + 3x_2 + x_3 = 6 \\ & x_1 \geq 0, \quad x_2 \geq 0 \quad \text{and} \quad x_3 \geq 0\end{array}$$

- Vertices correspond to choosing 2 of 3 variables to be zero
- Use the equation to solve for the value of the third variable



2 Basic feasible solutions

Basic feasible solutions (BFS) are a practical way of characterising a vertex, and I hope that the reason why they are needed will come through in what's set out below. The concept of a basic (feasible) solution will be the trickiest theoretical aspect of this course for many. Understanding it is a matter of getting the algebra and geometry clear in your head. It's worth it, as basic feasible solutions are the foundation of the simplex method and the properties of an optimal solution of an LP.

Each basic solution corresponds to a different way of choosing the m (basic) variables to solve for when considering the system of m equations $\bar{A}\mathbf{x} = \mathbf{b}$. This is equivalent to choosing n (nonbasic) variables to fix at zero so that there is still a solution of the system of equations - illustrated by the example in Lecture 3a with 3 variables and a single equation. There is a combinatorial number - $(m+n)!/(m!n!)$ - of choices of n from $n+m$, although not all yield basic solutions since the coefficient matrix of the system of equations corresponding to the remaining m variables may be singular. [As you know, three planes in $\mathbb{R}^3$ may meet in a line, rather than intersecting at a point.] As an illustration of this, look at the solution of Question 1 of Workshop 1 by vertex enumeration. When the point of intersection at a basic solution is feasible (is non-negative) then it is a basic feasible solution.

Since a vertex of the feasible region K is defined by n bound planes intersecting at a (feasible) point in the solution set of $\bar{A}\mathbf{x} = \mathbf{b}$, there's a basic feasible solution corresponding to every vertex of K . Conversely, every basic feasible solution corresponds to a vertex, since the n variables fixed at zero correspond to the n bound planes whose intersection defines a vertex.

Unfortunately, the vertices and BFS aren't in one-to-one correspondence: there may be more than one basic feasible solution at a vertex. This can happen if one of the basic variables is zero: there are then $n+1$ zero variables from which to choose a different set of n nonbasic variables, all at the same point in space. Such a situation is referred to as being **degenerate**, and is illustrated by the example in Question 3 of Workshop 2. Degeneracy isn't just a pathological situation invented by mathematicians: it's a feature of practical problems that has implications for solution methods.

At a basic feasible solution, the vector $\mathbf{x}$ of variables is partitioned into basic and nonbasic components $\mathbf{x}_B$ and $\mathbf{x}_N$. At the actual point corresponding to the basic feasible solution, the values of $\mathbf{x}_B$ and $\mathbf{x}_N$ are $\hat{\mathbf{b}}$ and $\mathbf{0}$.

However, for the purposes of analysis, we want to discuss general points in the feasible region K relative to the basic feasible solution. Relative to a basic feasible solution, any solution of $\bar{A}\mathbf{x} = \mathbf{b}$ is given by $\mathbf{x}_B = \hat{\mathbf{b}} - \hat{N}\mathbf{x}_N$ for some $\mathbf{x}_N \in \mathbb{R}^n$, where $\hat{N} = B^{-1}N$ and $\hat{\mathbf{b}} = B^{-1}\mathbf{b}$. The derivation of this is given in Lecture 3a, and comes from the equations $\bar{A}\mathbf{x} = \mathbf{b}$ being expressible as $B\mathbf{x}_B + N\mathbf{x}_N = \mathbf{b}$ and the matrix B being non-singular. For this point $\mathbf{x}$ to be in K , the values of $\mathbf{x}_N$ must be non-negative. Although $\mathbf{x}_B = \hat{\mathbf{b}} - \hat{N}\mathbf{x}_N$ is not in K for all $\mathbf{x}_N \geq \mathbf{0}$, all points in K are given by $\mathbf{x}_B = \hat{\mathbf{b}} - \hat{N}\mathbf{x}_N$ for *some* $\mathbf{x}_N \geq \mathbf{0}$.

Example 3 in Section 1

To illustrate expressing the basic variables in terms of the nonbasic variables, and a basic feasible solution, consider Example 3 in Section 1.

This has $m = 1$ and $n = 2$. Now, for (e.g.) $\mathcal{B} = \{3\}$, $\mathbf{x}_B = [x_3]$ and $B = [1]$ is nonsingular

- $\mathcal{N} = \{1, 2\}$, $N = \begin{bmatrix} 2 & 3 \end{bmatrix}$ and $\mathbf{x}_N = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$
- $\hat{\mathbf{b}} = [6]$ and $\hat{N} = \begin{bmatrix} 2 & 3 \end{bmatrix}$

Hence

$$\mathbf{x}_B = \hat{\mathbf{b}} - \hat{N}\mathbf{x}_N \quad \text{is} \quad x_3 = 6 - 2x_1 - 3x_2$$

for $x_1 \geq 0$ and $x_2 \geq 0$. Thus $\mathcal{B} = \{3\}$ corresponds to a basic feasible solution of $\bar{A}\mathbf{x} = \mathbf{b}$, with vertex $\mathbf{x} = [0 \ 0 \ 6]^T$ corresponding to $\mathbf{x}_N = \mathbf{0}$ and $\mathbf{x}_B = [6]$

The ScotFoods model (Workshop 1)

To illustrate expressing the basic variables in terms of the nonbasic variables, and a basic feasible solution, consider The ScotFoods model (Workshop 1).

This has $m = 3$ and $n = 2$

$$\begin{array}{llllllll} \max & f & = & 600x_1 & + & 900x_2 & & & \\ \text{s. t.} & & & 0.2x_1 & + & 0.5x_2 & + & x_3 & = & 200 \\ & & & 0.4x_1 & & & & + & x_4 & = & 180 \\ & & & 0.4x_1 & + & 0.5x_2 & & & + & x_5 & = & 250 \\ & & & x_1 \geq 0 & & x_2 \geq 0 & & x_3 \geq 0 & & x_4 \geq 0 & & x_5 \geq 0 \end{array}$$

Now, for (e.g.) $\mathcal{B} = \{1, 2, 4\}$, $\mathbf{x}_B = \begin{bmatrix} x_1 \\ x_2 \\ x_4 \end{bmatrix}$ and $B = \begin{bmatrix} 0.2 & 0.5 & 0 \\ 0.4 & 0 & 1 \\ 0.4 & 0.5 & 0 \end{bmatrix}$ is nonsingular. So

$$\bullet \mathcal{N} = \{3, 5\}, N = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \text{ and } \mathbf{x}_N = \begin{bmatrix} x_3 \\ x_5 \end{bmatrix}$$

$$\bullet \hat{\mathbf{b}} = B^{-1} \begin{bmatrix} 200 \\ 180 \\ 250 \end{bmatrix} = \begin{bmatrix} 250 \\ 300 \\ 80 \end{bmatrix} \text{ and } \hat{N} = \begin{bmatrix} -5 & 5 \\ 4 & -2 \\ 2 & -2 \end{bmatrix}$$

Hence

$$\mathbf{x}_B = \hat{\mathbf{b}} - \hat{N}\mathbf{x}_N \text{ is } \begin{bmatrix} 250 \\ 300 \\ 80 \end{bmatrix} - \begin{bmatrix} -5 & 5 \\ 4 & -2 \\ 2 & -2 \end{bmatrix} \mathbf{x}_N$$

for $x_1 \geq 0$ and $x_2 \geq 0$. Thus $\mathcal{B} = \{1, 2, 4\}$, corresponds to a basic feasible solution of $\bar{A}\mathbf{x} = \mathbf{b}$, with $\mathbf{x} = [250 \ 300 \ 0 \ 80 \ 0]^T$ corresponding to $\mathbf{x}_N = \mathbf{0}$ and $\mathbf{x}_B = \begin{bmatrix} 250 \\ 300 \\ 80 \end{bmatrix}$ being a feasible vertex of K

Example 4

As an example of a **partitioned LP**, basic feasible solution and reduced costs, consider the following LP in standard form

$$\begin{array}{llllllll} \text{maximize} & f & = & 2x_1 & + & x_2 & + & 3x_3 & & & \\ \text{subject to} & & & x_1 & & & + & 3x_3 & + & x_4 & = & 6 \\ & & & x_1 & + & 2x_2 & & & + & x_5 & = & 3 \\ & & & x_1 \geq 0, & x_2 \geq 0, & x_3 \geq 0, & x_4 \geq 0 & \text{ and } & x_5 \geq 0. \end{array}$$

has

$$\bar{\mathbf{c}} = [2 \ 1 \ 3 \ 0 \ 0]^T \text{ and } \bar{A} = \begin{bmatrix} 1 & 0 & 3 & 1 & 0 \\ 1 & 2 & 0 & 0 & 1 \end{bmatrix}$$

Corresponding to $\mathcal{B} = \{1, 3\}$, $\mathcal{N} = \{2, 4, 5\}$, the (identical) **partitioned LP** is

$$\begin{array}{llllllll} \text{maximize} & f & = & 2x_1 & + & 3x_3 & + & x_2 & & & \\ \text{subject to} & & & x_1 & + & 3x_3 & & + & x_4 & = & 6 \\ & & & x_1 & & + & 2x_2 & & + & x_5 & = & 3 \\ & & & x_1 \geq 0, & x_2 \geq 0, & x_3 \geq 0, & x_4 \geq 0 & \text{ and } & x_5 \geq 0. \end{array}$$

which can be written as

$$\begin{aligned} \text{maximize } f &= \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_2 \\ x_4 \\ x_5 \end{bmatrix} \\ \text{subject to } &\begin{bmatrix} 1 & 3 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_2 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix} \\ &\mathbf{x} \geq \mathbf{0} \end{aligned}$$

$\Longleftrightarrow$

$$\begin{aligned} \text{maximize } f &= \mathbf{c}_B^T \mathbf{x}_B + \mathbf{c}_N^T \mathbf{x}_N \\ \text{subject to } &B\mathbf{x}_B + N\mathbf{x}_N = \mathbf{b} \quad \mathbf{x} \geq \mathbf{0} \end{aligned}$$

for

$$B = \begin{bmatrix} 1 & 3 \\ 1 & 0 \end{bmatrix}, \quad N = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad \text{and} \quad \mathbf{c}_N = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Now

$$B\hat{\mathbf{b}} = \mathbf{b} \quad \Longleftrightarrow \quad \begin{bmatrix} 1 & 3 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix} \quad [\text{Solving for } \hat{b}_1 = 3/1 = 3 \text{ then } \hat{b}_2 = (6 - 3)/3 = 1]$$

corresponding to the feasible vertex $\mathbf{x} = [3 \ 0 \ 1 \ 0 \ 0]^T$, with reduced costs $\hat{\mathbf{c}}_N$ given by

$$\begin{aligned} B^T \boldsymbol{\pi} &= \mathbf{c}_B \quad \Longleftrightarrow \quad \begin{bmatrix} 1 & 1 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} \pi_1 \\ \pi_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad [\text{Solving for } \pi_1 = 3/3 = 1 \text{ then } \pi_2 = (2 - 1)/3 = 1] \\ \Rightarrow \quad \hat{\mathbf{c}}_N &= \mathbf{c}_N - N^T \boldsymbol{\pi} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 & 2 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix} \end{aligned}$$

3 Optimality conditions

Theorem 3 shows that, at a basic feasible solution $\mathbf{x}$, if the reduced costs are non-positive, then the basic feasible solution is optimal. Understanding why this is true is important for understanding the simplex method and properties of an optimal solution of an LP.

Theorem 3 A sufficient optimality condition for LP problems

A point $\mathbf{x} \in K$ is an optimal solution of an LP problem if it is a basic feasible solution with non-positive reduced costs $\hat{\mathbf{c}}_N = \mathbf{c}_N - N^T B^{-T} \mathbf{c}_B$

Proof:

- **With more words:**

The proof establishes the property that, at any point in the feasible region K other than this basic feasible solution $\mathbf{x}$, the objective function cannot be bigger than its value $\hat{f}$ at $\mathbf{x}$. To see why this is true, consider the following:

- Use $\mathbf{x}'$ to refer to any (other) point in K than basic feasible solution $\mathbf{x}$. Since both are feasible, $\overline{A}\mathbf{x} = \mathbf{b}$ and $\overline{A}\mathbf{x}' = \mathbf{b}$. So, subtracting, $\overline{A}\mathbf{d} = \mathbf{0}$, where $\mathbf{d} = \mathbf{x}' - \mathbf{x}$.
- Partitioning $\mathbf{d}$ in the same way as $\mathbf{x}$ into basic and nonbasic components $\mathbf{d}_B$ and $\mathbf{d}_N$, the values at the point $\mathbf{x}'$ can be partitioned as

$$\begin{bmatrix} \mathbf{x}_B \\ \mathbf{x}_N \end{bmatrix} = \begin{bmatrix} \hat{\mathbf{b}} \\ \mathbf{0} \end{bmatrix} + \begin{bmatrix} \mathbf{d}_B \\ \mathbf{d}_N \end{bmatrix}$$

Note that we also have $\mathbf{d}_N \geq \mathbf{0}$ since $\mathbf{x}' \in K$ so $\mathbf{x}_N \geq \mathbf{0}$.

- Partitioning $\overline{A}\mathbf{d} = \mathbf{0}$ similarly, we must have $B\mathbf{d}_B + N\mathbf{d}_N = \mathbf{0}$. Hence, since B is nonsingular, $\mathbf{d}_B = -B^{-1}N\mathbf{d}_N$.
- Now, the objective value at $\mathbf{x}'$ is $f = \mathbf{c}_B^T(\hat{\mathbf{b}} + \mathbf{d}_B) + \mathbf{c}_N^T\mathbf{d}_N$
- Substituting $\mathbf{d}_B = -B^{-1}N\mathbf{d}_N$ and simplifying, gives $f = \hat{f} + \hat{\mathbf{c}}_N^T\mathbf{d}_N$, where $\hat{\mathbf{c}}_N = \mathbf{c}_N - N^T B^{-T} \mathbf{c}_B$.
- Now, since $\mathbf{d}_N \geq \mathbf{0}$, $\hat{\mathbf{c}}_N \leq \mathbf{0}$ implies that $\hat{\mathbf{c}}_N^T\mathbf{d}_N \leq 0$, so $f \leq \hat{f}$.
- Hence, since the objective function at $\mathbf{x}'$ cannot be bigger than the $\hat{f}$ at the basic feasible solution, the latter must be an optimal solution of the LP.

- **Mathematically:**

Let $\mathbf{x}'$ be any point in K with $\mathbf{d} = \mathbf{x}' - \mathbf{x}$ so $\mathbf{x}' = \mathbf{x} + \mathbf{d}$ so $\mathbf{x}'$ is partitioned as

$$\begin{bmatrix} \mathbf{x}_B \\ \mathbf{x}_N \end{bmatrix} = \begin{bmatrix} \hat{\mathbf{b}} \\ \mathbf{0} \end{bmatrix} + \begin{bmatrix} \mathbf{d}_B \\ \mathbf{d}_N \end{bmatrix} \quad \text{with} \quad \mathbf{d}_N \geq \mathbf{0}$$

Now, at $\mathbf{x}'$, since $\overline{A}\mathbf{d} = \mathbf{0} \Rightarrow B\mathbf{d}_B + N\mathbf{d}_N = \mathbf{0} \Rightarrow \mathbf{d}_B = -B^{-1}N\mathbf{d}_N$

$$\begin{aligned} f &= \mathbf{c}_B^T(\hat{\mathbf{b}} + \mathbf{d}_B) + \mathbf{c}_N^T\mathbf{d}_N \\ &= \hat{f} - \mathbf{c}_B^T B^{-1}N\mathbf{d}_N + \mathbf{c}_N^T\mathbf{d}_N \quad [\text{Since } \overline{A}\mathbf{d} = \mathbf{0} \Rightarrow \mathbf{d}_B = -B^{-1}N\mathbf{d}_N] \\ &= \hat{f} + (\mathbf{c}_N - N^T B^{-T} \mathbf{c}_B)^T \mathbf{d}_N \\ &= \hat{f} + \hat{\mathbf{c}}_N^T \mathbf{d}_N \end{aligned}$$

Since $\mathbf{d}_N \geq \mathbf{0}$ it follows that $\hat{\mathbf{c}}_N \leq \mathbf{0} \Rightarrow \hat{\mathbf{c}}_N^T \mathbf{d}_N \leq 0 \Rightarrow f \leq \hat{f}$ so $\mathbf{x}$ is optimal.

4 The simplex algorithm

The simplex algorithm is one of the *Top Ten Algorithms of the 20th Century* for a reason. Invented in 1947 by George Dantzig (1914–2005), it's hugely important since it is the most widely-used technique for solving linear programming (LP) problems and is crucial for solving mixed-integer programming (MIP) problems. MIPs are LPs in which (some of) the variables have to take discrete (usually integer, and normally just binary) values. The ability to solve LPs and MIPs is critical to optimal decision-making throughout industry, the public sector, and the scientific world.

The simplex algorithm is also complicated, without being impossible to understand, and tests understanding of linear algebra and the geometry of planes in multiple dimensions. All together it's a great thing to teach to undergraduates! It was so badly taught when I first encountered it, that I ended up with little understanding of why it worked, but I could use it to solve LP problems with the “tableau simplex method”. This is a technique that's useless in practice, but taught to students because it can be learned without any understanding of what's actually going on mathematically: it's not what is taught in LPMS! I don't pretend that all members of the LPMS class will understand the simplex algorithm completely, but I've done my best to motivate it in a way that will give the class a better chance of doing so than I had!

The following description and definition are equivalent: the former is mainly words, the latter is mainly maths.

4.1 Description

Given a basic feasible solution $\mathbf{x}$

1. If the reduced costs are non-positive then **stop**
The solution is optimal
2. Determine the nonbasic variable $x_{q'}$ with most positive reduced cost
3. Determine the feasible direction $\mathbf{d}$ when $x_{q'}$ is increased from zero
4. If no basic variable is zeroed on $\mathbf{x} + \alpha\mathbf{d}$ then **stop**
The LP is unbounded
5. Determine the first basic variable $x_{p'}$ to be zeroed on $\mathbf{x} + \alpha\mathbf{d}$
6. Make $x_{p'}$ nonbasic and $x_{q'}$ basic
7. Go to 1

4.2 Definition

Given a basic feasible solution $\mathbf{x}$ with $\mathcal{B}$ and $\mathcal{N}$

1. If $\hat{\mathbf{c}}_{\mathcal{N}} \leq \mathbf{0}$ then **stop: the solution is optimal**
2. Determine the index $q' \in \mathcal{N}$ of the variable $x_{q'}$ with most positive reduced cost $\hat{c}_{q'}$
3. Let $\hat{\mathbf{a}}_q = B^{-1}\mathbf{a}_q$, where $\mathbf{a}_q$ is column q of N
4. If $\hat{\mathbf{a}}_q \leq \mathbf{0}$ then **stop: the LP is unbounded**
5. Determine the index $p' \in \mathcal{B}$ of the variable $x_{p'}$ corresponding to $p = \operatorname{argmin}_{\substack{i=1, \dots, m \\ \hat{a}_{iq} > 0}} \frac{\hat{b}_i}{\hat{a}_{iq}}$

6. Exchange indices p' and q' between $\mathcal{B}$ and $\mathcal{N}$ to yield a new basic feasible solution
7. Go to 1

4.3 Discussion

If the following discussion, there are some “Asides” that don’t have to be read, but should answer questions that you may have.

The simplex algorithm assumes that a basic feasible solution (which is also a vertex of the feasible region K) is known. How one can be found is considered later. Remember that, at a basic feasible solution, the indices $\{1, \dots, n+m\}$ are partitioned into sets $\mathcal{B}$ and $\mathcal{N}$ of indices of basic and nonbasic variables. The basic variables have value $\hat{\mathbf{b}} = B^{-1}\mathbf{b}$, and the nonbasic variables are zero. By calculating the reduced costs, Theorem 3 can be used to identify whether the basic feasible solution is optimal. If it is, then the simplex algorithm stops, because a solution to the LP has been found.

[Aside: If the solution is non-unique, then it’s possible to explore the set of solutions with simplex-like techniques, but it’s hardly ever done. If the basic feasible solution doesn’t have reduced costs satisfying $\hat{\mathbf{c}}_N \leq \mathbf{0}$ then the corresponding point may be optimal, but there’s no immediate way of proving it. Indeed, the simplex algorithm is a way of doing so.]

If the reduced costs aren’t non-positive, then there must be a positive reduced cost and, if there’s more than one, a most positive.

- The index q is used to refer to the index of the component $\hat{c}_q$ of $\hat{\mathbf{c}}_N$ that is the most positive.
- The index q' is used to refer to the q^{th} index of the set $\mathcal{N}$.

The simplex algorithm chooses to increase the nonbasic variable $x_{q'}$ from zero whilst keeping all the other nonbasic variables equal to zero, since feasible points when $x_{q'}$ is positive will have greater objective value than at the current basic feasible solution.

[Aside: When starting from an all-slack basis (see later) $\mathcal{N} = \{1, \dots, n\}$ so, in the first iteration q and q' are identical. This can be confusing and make students ask the question “Why do we need *both* q and q' !”. If I could do without one of them, believe me I would! However, in general, $\mathcal{N}$ isn’t $\{1, \dots, n\}$, and it’s necessary to distinguish the index q' of the variable chosen, and the position q where it is in $\mathcal{N}$, and where its reduced cost is in $\hat{\mathbf{c}}_N$.]

It’s not possible to increase $x_{q'}$ from zero without considering the values of the basic variables, since feasible points must satisfy $\overline{A}\mathbf{x} = \mathbf{b}$. Analysis performed earlier shows that if the change in the nonbasic variables is $\alpha\mathbf{d}_N$ (for $\alpha \geq 0$), then the change in the basic variables must be $\alpha\mathbf{d}_B = -\alpha B^{-1}N\mathbf{d}_N$, so that the equations remain satisfied along the line $\mathbf{x} + \alpha\mathbf{d}$. [It ensures that $\overline{A}\mathbf{d} = B\mathbf{d}_B + N\mathbf{d}_N = \mathbf{0}$.] The simplex algorithm chooses $\mathbf{d}_N = \mathbf{e}_q$ so the q^{th} nonbasic variable ($x_{q'}$) increases from zero to the value α , and all the other nonbasic variables remain zero. Hence, using $\mathbf{a}_q$ to denote $N\mathbf{e}_q$ (the q^{th} column of N), the basic variables must change by $\alpha\mathbf{d}_B = -\alpha B^{-1}\mathbf{a}_q = -\alpha\hat{\mathbf{a}}_q$, where $\hat{\mathbf{a}}_q = B^{-1}\mathbf{a}_q$. For this change in $x_{q'}$, the objective is $\hat{f} + \alpha\hat{c}_q > \hat{f}$, for $\alpha > 0$.

If all the components of $\hat{\mathbf{a}}_q$ are non-positive, then no basic variable reduces with α , so the objective can be increased (at rate $\hat{c}_q$) without bound by increasing $x_{q'}$. The LP is deduced as being unbounded.

Otherwise, the basic variables corresponding to positive components of $\hat{\mathbf{a}}_q$ are reduced as $x_{q'}$ is increased. For i such that $\hat{a}_{iq}$ (i^{th} component of $\hat{\mathbf{a}}_q$) is positive, basic variable i has value

$$\hat{b}_i - \alpha\hat{a}_{iq}, \quad \text{and is zeroed when} \quad \alpha_i = \frac{\hat{b}_i}{\hat{a}_{iq}}$$

There is a set of such “ratios”, and α must not be larger than the smallest of them, otherwise the point on $\mathbf{x} + \alpha\mathbf{d}$ is not feasible. The “ratio test”

$$p = \operatorname{argmin}_{i=1, \dots, m} \frac{\hat{b}_i}{\hat{a}_{iq} > 0}$$

finds the index p of the smallest ratio. The index p' is the p^{th} entry of $\mathcal{B}$. Basic variable $x_{p'}$ is zeroed, and will become nonbasic when indices p' and q' are interchanged between sets $\mathcal{B}$ and $\mathcal{N}$ as the last step in the simplex algorithm.

Geometrically, the n variables corresponding to the indices in $\mathcal{N}$ are zero at the basic feasible solution, and they define a vertex of the feasible region K . Of these n nonbasic variables, $n - 1$ remain zero as $x_{q'}$ is increased from zero. The line of points $\mathbf{x} + \alpha \mathbf{d}$ as $x_{q'}$ is increased is an edge of the feasible region. Unless unboundedness is detected, this edge ends when it reaches a new vertex of K : when the first of the variables that was basic at the the previous vertex is zeroed.

4.4 An initial basic feasible solution

In exercises you'll be given an initial basic feasible solution. However, in practice, the first one to try is the “all slack” basis $\mathcal{B} = \{n + 1, \dots, n + m\}$ and $\mathcal{N} = \{1, \dots, n\}$ in which the slack variables are basic, and the original variables are nonbasic:

$$\mathbf{x}_B = \begin{bmatrix} x_{n+1} \\ \vdots \\ x_{n+m} \end{bmatrix} \quad \text{and} \quad \mathbf{x}_N = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}.$$

This has the property that the basis matrix $B = I$ is non-singular, so it's a basic solution. Since the values of the basic variables are $\hat{\mathbf{b}} = B^{-1}\mathbf{b} = \mathbf{b}$, it's a basic feasible solution if $\mathbf{b} \geq \mathbf{0}$. If this does not hold, then a modified LP problem is solved (see later).

4.5 Worked example

For the following LP in standard form

$$\begin{aligned} \text{maximize } f &= 4x_1 + x_2 + 6x_3 \\ \text{subject to } & x_1 + 3x_3 + x_4 = 6 \\ & x_1 + 2x_2 + x_5 = 3 \\ & x_1 \geq 0 \quad x_2 \geq 0 \quad x_3 \geq 0 \quad x_4 \geq 0 \quad x_5 \geq 0, \end{aligned}$$

- RHS $\mathbf{b} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$ is positive so the “all-slack” basis $\mathcal{B} = \{4, 5\}$ and $\mathcal{N} = \{1, 2, 3\}$ yields a basic feasible solution

- **Iteration 1**

- For $\mathcal{B} = \{4, 5\}$ and $\mathcal{N} = \{1, 2, 3\}$, $B = I$, $N = \begin{bmatrix} 1 & 0 & 3 \\ 1 & 2 & 0 \end{bmatrix}$, $\mathbf{c}_B = \mathbf{0}$ and $\mathbf{c}_N = \begin{bmatrix} 4 \\ 1 \\ 6 \end{bmatrix}$, giving

$$\hat{\mathbf{b}} = B^{-1}\mathbf{b} = I^{-1} \begin{bmatrix} 6 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

- $\hat{\mathbf{c}}_N = \mathbf{c}_N - N^T B^{-T} \mathbf{c}_B = \begin{bmatrix} 4 \\ 1 \\ 6 \end{bmatrix} - N^T B^{-T} \mathbf{0} = \begin{bmatrix} 4 \\ 1 \\ 6 \end{bmatrix}$.

- Choose $q = 3$ since $\hat{c}_q = 6$ is most positive, and hence $q' = 3$ [q^{th} index in $\mathcal{N}$]

- Form $\hat{\mathbf{a}}_q$ as solution of $B\hat{\mathbf{a}}_q = \mathbf{a}_q$ for $\mathbf{a}_q = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$, being column $q = 3$ of N .

$$\text{Since } B = I, \hat{\mathbf{a}}_q = I^{-1} \begin{bmatrix} 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

- Choose

$$p = \operatorname{argmin} \left\{ \begin{array}{c} \frac{6}{3} \\ (i = 1) \end{array} \right\} = 1$$

so $p' = 4$ [p^{th} index in $\mathcal{B}$]

- Interchange $q' = 3$ and $p' = 4$ between $\mathcal{B}$ and $\mathcal{N}$ and the new basic feasible solution is given by $\mathcal{B} = \{3, 5\}$ and $\mathcal{N} = \{1, 2, 4\}$.

• Iteration 2

- For $\mathcal{B} = \{3, 5\}$ and $\mathcal{N} = \{1, 2, 4\}$, $B = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$, $N = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 2 & 0 \end{bmatrix}$, $\mathbf{c}_B = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$ and $\mathbf{c}_N = \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}$,
giving $\hat{\mathbf{b}}$ as the solution of $B\hat{\mathbf{b}} = \mathbf{b}$ which, by inspection, is $\begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$ so $\hat{\mathbf{b}} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

- $\hat{\mathbf{c}}_N = \mathbf{c}_N - N^T B^{-T} \mathbf{c}_B = \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix} - N^T B^{-T} \begin{bmatrix} 6 \\ 0 \end{bmatrix}$. Now $B^{-T} \begin{bmatrix} 6 \\ 0 \end{bmatrix}$ is the solution of $B^T \boldsymbol{\pi} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$

which, by inspection, is $\begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$ so $\hat{\mathbf{c}}_N = \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$

- Choose $q = 1$ since $\hat{c}_q = 2$ is most positive, and hence $q' = 1$ [q^{th} index in $\mathcal{N}$]
- Form $\hat{\mathbf{a}}_q$ as solution of $B\hat{\mathbf{a}}_q = \mathbf{a}_q$ for $\mathbf{a}_q = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, being column $q = 1$ of N .

By inspection, $B\hat{\mathbf{a}}_q = \mathbf{a}_q$ is $\begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1/3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ so $\hat{\mathbf{a}}_q = \begin{bmatrix} 1/3 \\ 1 \end{bmatrix}$

- Choose

$$p = \operatorname{argmin} \left\{ \begin{array}{c} \frac{2}{1/3}, \frac{3}{1} \\ (i = 1), (i = 2) \end{array} \right\} = \operatorname{argmin} \left\{ \begin{array}{c} 6, 3 \\ (i = 1), (i = 2) \end{array} \right\} = 2$$

so $p' = 5$ [p^{th} index in $\mathcal{B}$]

- Interchange $q' = 1$ and $p' = 5$ between $\mathcal{B}$ and $\mathcal{N}$ and the new basic feasible solution is given by $\mathcal{B} = \{3, 1\}$ and $\mathcal{N} = \{5, 2, 4\}$.

[The order of indices within $\mathcal{B}$ and (particularly) $\mathcal{N}$ is relatively unimportant, so long as you are consistent with the ordering throughout a particular simplex iteration.]

• Iteration 3

- For $\mathcal{B} = \{3, 1\}$ and $\mathcal{N} = \{5, 2, 4\}$, $B = \begin{bmatrix} 3 & 1 \\ 0 & 1 \end{bmatrix}$, $N = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 2 & 0 \end{bmatrix}$, $\mathbf{c}_B = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$ and $\mathbf{c}_N = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$,
giving $\hat{\mathbf{b}}$ as the solution of $B\hat{\mathbf{b}} = \mathbf{b}$ which, by inspection as a triangular solve, is $\begin{bmatrix} 3 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$ so $\hat{\mathbf{b}} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

$$\circ \hat{\mathbf{c}}_N = \mathbf{c}_N - N^T B^{-T} \mathbf{c}_B = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - N^T B^{-T} \begin{bmatrix} 6 \\ 4 \end{bmatrix}. \text{ Now } B^{-T} \begin{bmatrix} 6 \\ 4 \end{bmatrix} \text{ is the solution of } B^T \boldsymbol{\pi} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

$$\text{which, by inspection as a triangular solve, is } \begin{bmatrix} 3 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix} \text{ so } \hat{\mathbf{c}}_N = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ 0 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} =$$

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ -3 \\ -2 \end{bmatrix}$$

◦ Since $\hat{\mathbf{c}}_N \leq \mathbf{0}$ satisfies the optimality condition **stop**.

Since $\begin{bmatrix} x_3 \\ x_1 \end{bmatrix} = \mathbf{x}_B = \hat{\mathbf{b}} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$, the optimal solution is $\mathbf{x} = [3 \ 0 \ 1 \ 0 \ 0]^T$, where values of x_2 , x_4 and x_5 are zero since $\mathcal{N} = \{5, 2, 4\}$.

5 The simplex algorithm: Issues

Sections 5 and 6 discuss a range of issues raised by the simplex algorithm. Some consist of analytical questions that should be addressed whenever any algorithm is studied, and others are specific.

5.1 Phase I: how to start if $\mathbf{b} \not\geq \mathbf{0}$

With the “all slack” basis

$$\mathcal{B} = \{n+1, \dots, n+m\} \quad \text{and} \quad \mathcal{N} = \{1, \dots, n\}$$

, the basis matrix $B = I$, so the values of the basic variables are $\hat{\mathbf{b}} = B^{-1}\mathbf{b} = \mathbf{b}$. However, the basis is **not** feasible if $\mathbf{b} \not\geq \mathbf{0}$.

To get around this situation, the simplex method is used to find a basic feasible solution, or prove that the LP is infeasible. The latter is a possibility that is known to exist, so any solution technique must accommodate it.

Since the “all slack” basis has some slack variables which take negative values, **artificial variables** are introduced in order to permit a basic feasible solution to be constructed. Remembering that slack variables measure the extent to which the original inequalities are satisfied, artificial variables measure the extent to which they are not. The **Phase I** procedure uses the simplex algorithm to minimize these “infeasibilities”, and exactly one of the following occurs

- The sum of infeasibilities is reduced to zero, so the simplex basis is feasible
- The sum of infeasibilities cannot be reduced to zero, so the LP is infeasible

For all $i = 1, \dots, m$, an artificial variable $x_{n+m+i} \geq 0$ is subtracted from each constraint i , and the original objective $f = \bar{\mathbf{c}}^T \mathbf{x}$ is replaced by the **Phase I** objective $f = -\sum_{i=1}^m x_{n+m+i}$, yielding the **Phase I problem**

$$\begin{array}{llllll} \max & f = & & & -x_{n+m+1} & \dots & -x_{n+2m} \\ \text{s. t.} & a_{11}x_1 & + & \dots & + & a_{1n}x_n & +x_{n+1} & & -x_{n+m+1} & & & = & b_1 \\ & & & \vdots & & \vdots & & \ddots & & \ddots & & \vdots \\ & a_{m1}x_1 & + & \dots & + & a_{mn}x_n & & +x_{n+m} & & & -x_{n+2m} & = & b_m \\ & x_1 \geq 0, & \dots, & x_{n+2m} \geq 0 & & & & & & & & \end{array}$$

An initial basic feasible solution is constructed as follows: For $i = 1, \dots, m$

- If $b_i \geq 0$: Slack $x_{n+i} = b_i \geq 0$ is basic; Artificial $x_{n+m+i} = 0$ is nonbasic; Column i of B is $\mathbf{e}_i$
- If $b_i < 0$: Slack x_{n+i} is nonbasic; Artificial $x_{n+m+i} = -b_i > 0$ is basic; Column i of B is $-\mathbf{e}_i$

The basis matrix B is nonsingular and, by construction, $\hat{\mathbf{b}} \geq \mathbf{0}$.

The Phase I objective is the negated sum of artificial variables, and its value is the negated sum of infeasibilities. Since the artificial variables are non-negative, and the Phase I objective is their negated sum, it cannot take a positive value. It is negative iff at least one artificial variable is positive. Hence the simplex method drives the Phase I objective up towards zero.

- If the Phase I objective reaches zero, then all artificial variables are zero. Since the corresponding basis has all original and slack variables either positive or zero, it is a basic feasible solution for the original LP problem (in standard form). The simplex algorithm is then applied to this (**Phase II**) problem, starting from the basic feasible solution obtained in Phase I.
- If the simplex algorithm identifies optimality for a negative value of the Phase I objective, then it's not possible for it to be increased to zero: it's not possible to find a basis with all artificial variables being zero. Hence the LP is deduced to be infeasible.

Example

For the LP

$$\begin{aligned} \text{maximize } f &= -x_1 - 2x_2 \\ \text{subject to } & -x_1 - 2x_2 \leq -4 \\ & -x_1 + 2x_2 \leq 2 \\ & x_1 \geq 0 \quad x_2 \geq 0, \end{aligned}$$

in standard form it is

$$\begin{aligned} \text{maximize } f &= -x_1 - 2x_2 \\ \text{subject to } & -x_1 - 2x_2 + x_3 = -4 \\ & -x_1 + 2x_2 + x_4 = 2 \\ & x_1 \geq 0 \quad x_2 \geq 0 \quad x_3 \geq 0 \quad x_4 \geq 0. \end{aligned}$$

Add artificial variables and form the Phase I problem

$$\begin{aligned} \text{maximize } f &= -x_5 - x_6 \\ \text{subject to } & -x_1 - 2x_2 + x_3 - x_5 = -4 \\ & -x_1 + 2x_2 + x_4 - x_6 = 2 \\ & x_1 \geq 0 \quad x_2 \geq 0 \quad x_3 \geq 0 \quad x_4 \geq 0 \quad x_5 \geq 0 \quad x_6 \geq 0. \end{aligned}$$

For the initial basis, choose slack or artificial variables to be basic according to the sign of b_i , $i = 1, \dots, m$.

- $b_1 < 0$ so x_5 is basic with value $-b_1 = 4$; x_3 is nonbasic
- $b_2 > 0$ so x_4 is basic with value $b_2 = 2$; x_6 is nonbasic

Simplex algorithm, from $\mathcal{B} = \{5, 4\}$ and $\mathcal{N} = \{1, 2, 3, 6\}$. Since $x_1 = x_2 = 0$, this is vertex O in the graphical representation. This point is visibly infeasible for the original LP since it violates the first constraint - which is why its slack variable (x_3) would have a negative value if it were basic.

Iteration 1

$$B = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \quad N = \begin{bmatrix} -1 & -2 & 1 & 0 \\ -1 & 2 & 0 & -1 \end{bmatrix} \quad \hat{\mathbf{b}} = B^{-1}\mathbf{b} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

Now

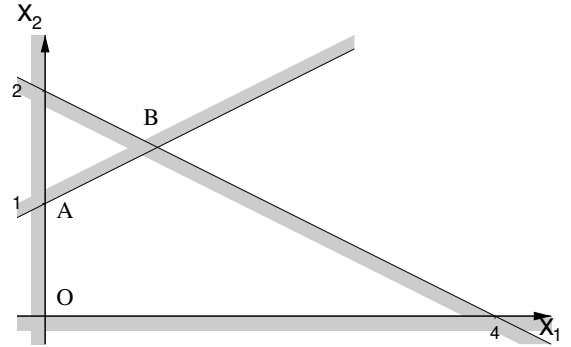
$$\mathbf{c}_B = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \quad \mathbf{c}_N = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -1 \end{bmatrix} \quad \Rightarrow \quad \hat{\mathbf{c}}_N = \begin{bmatrix} 1 \\ 2 \\ -1 \\ -1 \end{bmatrix}$$

From the image, it can be seen that increasing x_1 or x_2 will reduce the sum of infeasibilities (which is just the violation of the first constraint). The most positive reduced cost has index $q = 2$, for which $q' = 2$. Computing

$$\hat{\mathbf{a}}_q = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \quad p = \operatorname{argmin} \left\{ \begin{array}{c} \frac{4}{2} \\ (i=1) \end{array}, \begin{array}{c} \frac{2}{2} \\ (i=2) \end{array} \right\} = 2 \quad \Rightarrow \quad p' = 4$$

This corresponds to increasing x_2 to 1: moving to the vertex A in the image. Interchange 2 and 4 between $\mathcal{B}$ and $\mathcal{N}$ to give $\mathcal{B} = \{2, 5\}$ and $\mathcal{N} = \{1, 3, 4, 6\}$.

Iteration 2



$$B = \begin{bmatrix} -2 & -1 \\ 2 & 0 \end{bmatrix} \quad N = \begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & -1 \end{bmatrix} \quad \hat{\mathbf{b}} = B^{-1}\mathbf{b} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Now

$$\mathbf{c}_B = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \quad \mathbf{c}_N = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -1 \end{bmatrix} \quad \Rightarrow \quad \hat{\mathbf{c}}_N = \begin{bmatrix} 2 \\ -1 \\ -1 \\ 0 \end{bmatrix}$$

From the image, it can be seen that increasing x_1 (in moving from A towards B , since x_4 remains zero) will further reduce the violation of the first constraint. The positive reduced cost has index $q = 1$, for which $q' = 1$. Computing

$$\hat{\mathbf{a}}_q = \begin{bmatrix} -1/2 \\ 2 \end{bmatrix}, \quad p = \operatorname{argmin}_{(i=2)} \left\{ \begin{array}{c} 2 \\ \frac{2}{2} \end{array} \right\} = 2 \quad \Rightarrow \quad p' = 5$$

This corresponds to increasing x_1 to 1: moving to the vertex B in the image. Interchange 1 and 5 between $\mathcal{B}$ and $\mathcal{N}$ to give $\mathcal{B} = \{1, 2\}$ and $\mathcal{N} = \{3, 4, 5, 6\}$.

Since the artificial variables (x_5 and x_6) are both nonbasic, they are zero so Phase I is complete. This is also “visible” in the image. The artificials can be removed from the problem, and the Phase II LP (in standard form with the original objective) can be solved from the basic feasible solution $\mathcal{B} = \{1, 2\}$ and $\mathcal{N} = \{3, 4\}$ [Exercise].

5.2 Correctness

For any algorithm, it’s important to ensure that it is well-defined. The discussion above has established that, given a basic feasible solution, all the steps in the simplex algorithm are well-defined, so what remains is to show that interchanging indices p' and q' between sets $\mathcal{B}$ and $\mathcal{N}$ yields a basic feasible solution so that the next iteration, and subsequent iterations, are well defined.

What is required, is to show that

- The new sets $\mathcal{B}$ and $\mathcal{N}$ form a partition of $\{1, 2, \dots, n + m\}$
 - Interchanging $p' \in \mathcal{B}$ and $q' \in \mathcal{N}$ between $\mathcal{B}$ and $\mathcal{N}$ clearly yields a partition of $\{1, 2, \dots, n+m\}$
- The nonbasic variables $\mathbf{x}_N = \mathbf{0}$
 - Only $x_{q'}$ is increased from zero and q' leaves $\mathcal{N}$; Other nonbasic variables remain zero; $x_{p'}$ is decreased to zero and p' enters $\mathcal{N}$
- The basic variables $\mathbf{x}_B = \hat{\mathbf{b}} \geq \mathbf{0}$
 - Since $x_{p'}$ was the *first* basic variable to be zeroed on $\mathbf{x} + \alpha \mathbf{d}$, all others are non-negative; $x_{q'} \geq 0$
- The new basis matrix B is nonsingular

Establishing this last property is non-trivial, but doing so requires some neat linear algebra and also feeds into methods for implementing the simplex algorithm practically.

Since the new $\mathcal{B}$ is the old $\mathcal{B}$ with entry p replaced by the q^{th} index from $\mathcal{N}$, the new basis matrix is the old basis matrix with column p replaced by the vector $\mathbf{a}_q$, which is column q of N .

Given this, the first thing to do is to establish the following, as an expression for the new basis matrix

$$B + (\mathbf{a}_q - B\mathbf{e}_p)\mathbf{e}_p^T$$

Since column j of any matrix M is Me_j , where e_j is column j of the appropriately-dimensioned identity matrix, this can be used to verify that the expression gives the new basis matrix. Noting that $e_p^T e_j$ is zero if $p \neq j$ and one if $p = j$, it follows that

$$[B + (\mathbf{a}_q - Be_p)e_p^T] e_j = \begin{cases} Be_j & j \neq p \\ \mathbf{a}_q & j = p \end{cases} \quad \begin{matrix} \text{[Same as column } j \text{ of } B] \\ \text{[New column } p \text{ of } B] \end{matrix}$$

This expression for the updated basis matrix is very important when implementing the simplex algorithm, as it allows the updated matrix to be written as

$$B + (\mathbf{a}_q - Be_p)e_p^T = B[I + (\hat{\mathbf{a}}_q - e_p)e_p^T] = BE$$

where $E = I + (\hat{\mathbf{a}}_q - e_p)e_p^T$. The matrix E performs what's known as a *rank-one* update of B , since $\mathbf{u}\mathbf{v}^T$ is a matrix of rank one so long as $\mathbf{u}$ and $\mathbf{v}$ are nonzero. Note that $\mathbf{u}\mathbf{v}^T$ is a matrix, but $\mathbf{v}^T\mathbf{u}$ is a scalar. Now, since B is nonsingular (so $\det(B) \neq 0$) and $\det(UV) = \det(U)\det(V)$ for any two square matrices, the updated basis matrix is nonsingular iff E is nonsingular. In general, it can be shown that E is nonsingular iff $\hat{a}_{pq}$ is nonzero, but $\hat{a}_{pq} > 0$ is guaranteed by the simplex algorithm since it's a denominator in the ratio test, and only positive denominators are considered. Hence the updated basis matrix B is nonsingular.

5.3 Termination

Having shown that an algorithm is correct, the next things to consider are: “Does it terminate?” and, if it does, “What is its computational complexity?”. So long as one particular situation doesn't occur, it's relatively easy to show that the simplex algorithm terminates.

Degeneracy

For feasibility, the value of the basic variables must satisfy $\hat{\mathbf{b}} \geq \mathbf{0}$. With one exception, in all of the examples in LPMS the property $\hat{\mathbf{b}} > \mathbf{0}$ holds: all the basic variables are positive. However, it is perfectly possible for a basic variable to have zero value. Consider the example

$$\begin{aligned} \text{maximize } f &= 4x_1 + 2x_2 + 0.1x_3 \\ \text{subject to } & \quad x_1 + x_2 \leq 0 \\ & x_1 \geq 0 \quad x_2 \geq 0 \quad x_3 \geq 0. \end{aligned}$$

Adding the slack variable x_4 , and starting from a slack basis ($\mathcal{B} = \{4\}$), $\hat{\mathbf{b}} = [0]$ so all variables are zero. Thus, when trying to increase x_1 from zero, a zero step is taken and the simplex algorithm stays at the origin. In the next iteration, when trying to increase x_2 from zero with x_1 basic, the same thing happens. There are three basic feasible solutions at the origin, corresponding to $\mathcal{B} = \{4\}$, $\mathcal{B} = \{1\}$ and $\mathcal{B} = \{2\}$ respectively. [In the third iteration, when increasing the nonbasic variable (x_3) that was least attractive in the first iteration, $\hat{\mathbf{a}}_q = [0]$ so the LP is identified as being unbounded.] When $\hat{\mathbf{b}}$ contains zero entries, the vertex is said to be **degenerate**, and the possibility of not making a positive step at a degenerate vertex prevents a simple proof of termination for general LPs. Indeed, failure to move from a degenerate vertex can lead to the simplex algorithm failing completely.

Theorem 4: Termination of the simplex algorithm in the absence of degeneracy

At the beginning of an iteration, if $\hat{\mathbf{c}}_N \leq \mathbf{0}$ then the algorithm terminates, and the basic feasible solution corresponds to an optimal solution. Otherwise, there is a chosen reduced cost $\hat{c}_q > 0$.

If $\hat{\mathbf{a}}_q \leq \mathbf{0}$ then the algorithm terminates because the LP is unbounded

Otherwise, the absence of degeneracy implies that $\hat{\mathbf{b}} > \mathbf{0}$ so $\bar{\alpha} = \frac{\hat{b}_p}{\hat{a}_{pq}} > 0$

So, since $\hat{c}_q > 0$, the objective increases by $\bar{\alpha}\hat{c}_q > 0$ in moving from the basic feasible solution at one vertex to the basic feasible solution at a different vertex.

This strict increase in the objective function means that the algorithm cannot return to a basic feasible solution previously visited.

There is a finite number of basic feasible solutions, so the algorithm must eventually terminate—either an optimal solution or by identifying unboundedness

5.4 Complexity

When an algorithm is guaranteed to terminate, the next piece of analysis to consider is the algorithm's (worst case) complexity. The proof of Theorem 4 relies on there being a finite number of feasible vertices, but leaves open the possibility that all feasible vertices can be visited. Does this ever happen?

The answer is that it can, and it does for the Klee-Minty problems. This is a family of LP problems derived by Klee and Minty (1971).

$$\begin{aligned} & \text{maximize} && \sum_{j=1}^n 10^{j-1} x_j \\ & \text{subject to} && x_i + 2 \sum_{j=i+1}^n 10^{j-i} x_j \leq 100^{n-i} \quad \text{for } i = 1, \dots, n \\ & && x_1 \geq 0, \dots, x_n \geq 0 \end{aligned}$$

The problems have n variables and n constraints, the feasible region has 2^n vertices, and the simplex algorithm visits all of them!

This upper bound of 2^n on number of iterations is **exponential** in the problem size. It has been shown that it is possible to solve LPs in a time that is **polynomial** in the problem size, and this is achieved by a class of (interior point) solution methods. So, why use the simplex method? Well, the *typical* number of iterations is $O(n+m)$, and the simplex method yields a basic feasible solution (unlike interior point methods. What this observation also shows is that results from complexity analysis may not be meaningful!

5.5 Cycling

As the example above illustrates, degeneracy can lead to the simplex algorithm failing to move from a degenerate vertex for several iterations. As the following example illustrates, in the worst case it can lead to the simplex algorithm failing completely due to it **cycling** through a sequence of basic feasible solutions that repeats indefinitely.

$$\begin{aligned} & \text{maximize} && x_1 - 5.5x_2 + 0.75x_3 - 5.75x_4 \\ & \text{subject to} && 2.5x_1 - 19.5x_2 - 3.5x_3 + 19.5x_4 + x_5 = 0 \\ & && 0.5x_1 - 3.5x_2 - 0.5x_3 + 2.5x_4 + x_6 = 0 \quad \mathbf{x} \geq \mathbf{0} \end{aligned}$$

This is one of a family of problems that was derived by Hall and McKinnon (1996), and has the “nicest” numbers! They are unbounded but, when started from a slack basis, the simplex algorithm cycles through a sequence of six basic feasible solutions without leaving the origin. They can also be shown to be the simplest LP cycling examples.

Although the Hall-McKinnon examples are pathological, and such exact cycling is rarely seen in practice, degeneracy occurs naturally in practical problems. Modifications to the simplex algorithm allow its termination for general LPs to be proved, but these techniques are too expensive to use in practice. Efficient, and more practical techniques exist that greatly reduce the consequences of degeneracy.

6 The simplex algorithm: Implementation

Hitherto, the simplex algorithm has been illustrated using “toy” LPs ($m \leq 3$, $n \leq 5$) on paper, with *ad hoc* calculations performed to obtain the necessary data. Sometimes (triangular) systems have been “by inspection” and, otherwise, B^{-1} has been formed explicitly. To solve practical LP problems requires the simplex algorithm to be applied systematically and efficiently. The **simplex method** is the algorithm and a systematic method of obtaining the data. When coded on a computer it becomes an **implementation**.

The data required by an iteration of the simplex algorithm, are the reduced RHS $\hat{\mathbf{b}}$, the reduced costs $\hat{\mathbf{c}}_N$, and the pivotal column $\hat{\mathbf{a}}_q$. The **revised simplex method** obtains these values as follows

$$\begin{array}{llll} \text{Solve} & B\hat{\mathbf{b}} & = & \mathbf{b} & \text{to obtain } \hat{\mathbf{b}} = B^{-1}\mathbf{b} \\ \text{Solve} & B^T\boldsymbol{\pi} & = & \mathbf{c}_B & \text{to obtain } \boldsymbol{\pi} = B^{-T}\mathbf{c}_B \\ \text{Form} & \hat{\mathbf{c}}_N & = & \mathbf{c}_N - N^T\boldsymbol{\pi} \\ \text{Solve} & B\hat{\mathbf{a}}_q & = & \mathbf{a}_q & \text{to obtain } \hat{\mathbf{a}}_q = B^{-1}\mathbf{a}_q \end{array}$$

Thus, a key requirement is to be able to solve efficiently systems of equations with coefficient matrix B (or B^T). The key properties of B that inform this computational requirement are that it is nonsingular, typically **sparse** and successive basis matrices are related by $B := B + (\mathbf{a}_q - B\mathbf{e}_p)\mathbf{e}_p^T$.

Sparse LP problems

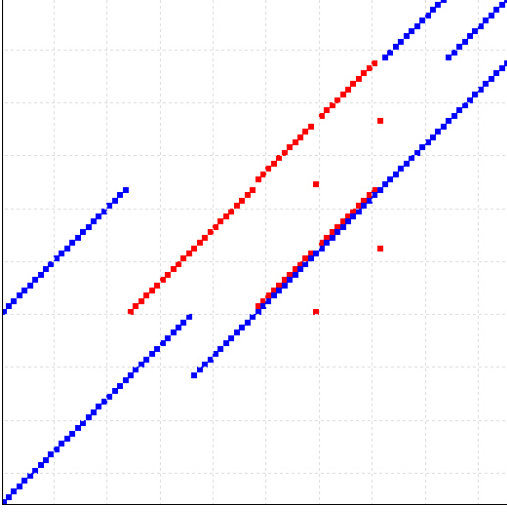
The entries of the coefficient matrix of the LP problems solved so far on paper have generally been nonzero. However, most constraint matrix coefficients of practical LP problems are zero. This is because each variable is usually involved in very few constraints. It is important for computational efficiency that the existence of these zero entries is exploited: additions and multiplications with zero should be avoided. The LP problem

$$\text{maximize } \mathbf{c}^T \mathbf{x} \quad \text{subject to } A\mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0}$$

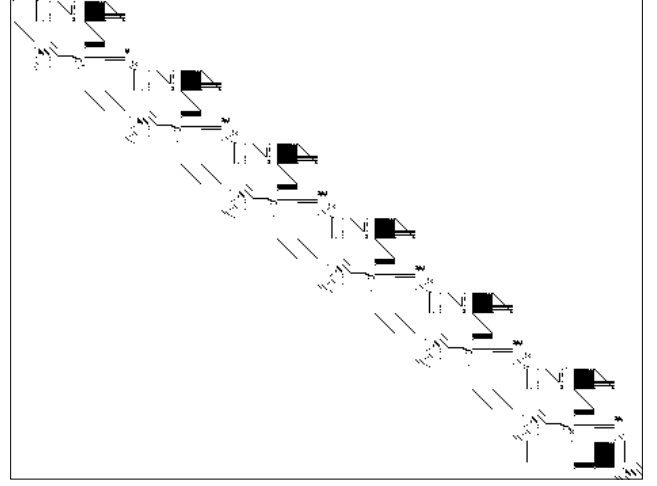
is **sparse** if zero values in A are worth exploiting computationally. If A is sparse then, since the basis matrix B is a square submatrix of $[A \ I]$ it is similarly sparse. The matrix may also have **structure** that is worth exploiting. The following two examples illustrate the constraint matrix for sparse LP problems: Figure 1a indicates the location of the nonzero values of the constraint matrix (its **sparsity pattern**), with the negative values being red, and the positive values, blue. Figure 1b gives the sparsity pattern of the constraint matrix for the (small) LP test problem STAIR, where some matrix structure is also seen. The number of nonzero entries per column is small for both problems and, such its the nature of modelling, the typical number of nonzeros per column doesn’t increase much whatever the dimension of an LP problem.

6.1 Inverting the matrix B

A single system of equations $B\mathbf{x} = \mathbf{b}$ can be solved using Gaussian elimination. However, when multiple systems involving B and its transpose have to be solved, it is inefficient to solve each of them using Gaussian elimination. What is more efficient is to form an **invertible representation** of B , and use it to solve the system directly. When not exploiting sparsity - or when the matrix B is not sparse - Gaussian elimination has a cost proportional to m^3 . However, once an invertible representation has been formed (at a cost proportional to m^3), the cost of using it to solving a single system is only proportional to m^2 . When m is large, this makes a *big* difference.



(a) Constraint matrix A for SLATS4U: $n = 96$, $m = 96$, $\tau = 216$, $\tau/n = 2.3$ nonzeros/column, density $100 \times \frac{216}{96 \times 96} = 2.3\%$



(b) Constraint matrix A for STAIR: $n = 467$, $m = 363$, $\tau = 3856$, $\tau/n = 8.3$ nonzeros/column, density $100 \times \frac{3856}{467 \times 363} = 2.3\%$

Figure 1: Sparsity patterns for constraint matrices of LPs with n columns (variables), m rows (constraints) and τ nonzeros

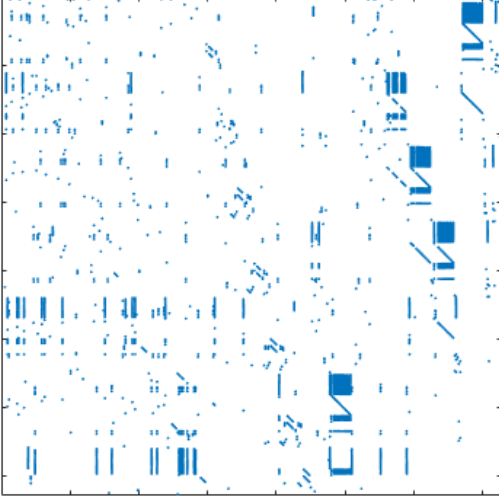
Why not form B^{-1} ?

The inverse is the most obvious invertible representation of a matrix, and can be formed in many ways. One of the least inefficient approaches is to solve m systems of equations by applying Gaussian elimination to the augmented matrix $[B \ I]$, reducing the B to I . This has a cost proportional to m^3 . Once B^{-1} is formed, using it to calculate $\mathbf{x} = B^{-1}\mathbf{b}$ by multiplying $\mathbf{b}$ has a cost proportional to m^2 , since every entry in B^{-1} is used exactly once. This latter observation is the key to finding and using efficient invertible representations.

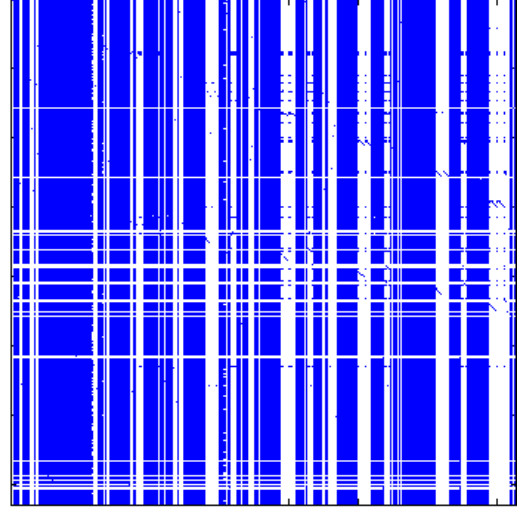
When B is a sparse matrix, it is unsurprising that B^{-1} is very much less so, since column j of the inverse matrix is the solution of the system of equations $B\mathbf{x}_j = \mathbf{e}_j$. This is illustrated by the sparsity patterns in Figure 2, where the density of B^{-1} is 61% for a matrix B of density 2.5%. This Figure also gives the **fill factor**, being the ratio between the number of nonzeros in the invertible representation and the number of nonzeros in the matrix. Further examples put this measure into context.

The LU decomposition

When traditional Gaussian elimination is used to solve $B\mathbf{x} = \mathbf{b}$, row operations are applied to the augmented matrix $[B \ \mathbf{b}]$ until it is reduced to upper triangular form $[U \ \mathbf{y}]$. The solution $\mathbf{x}$ can then be found by solving $U\mathbf{x} = \mathbf{y}$ via **backward substitution**. The first value to be solved for is $x_m = y_m/u_{mm}$ and, after this is substituted into equation $m - 1$, $x_{m-1} = (y_m - x_m u_{m-1m})/u_{m-1m-1}$, etc. When performing Gaussian elimination, a multiplier is calculated for each entry below the diagonal to be eliminated. The matrix L of these multipliers (in the locations corresponding to the entries they eliminate), with unit entries on the diagonal, can be shown to satisfy $B = LU$, where U is the upper triangular matrix obtained by Gaussian elimination. Since the LU decomposition can be used to solve $B\mathbf{x} = \mathbf{b}$ without any further elimination, it is an invertible representation of B .



(a) B has 3279 nonzeros and density 2.5%



(b) B^{-1} has 80284 nonzeros, density 61% and fill factor $\frac{80284}{3279} = 24.5$

Figure 2: Forming the inverse of B for STAIR

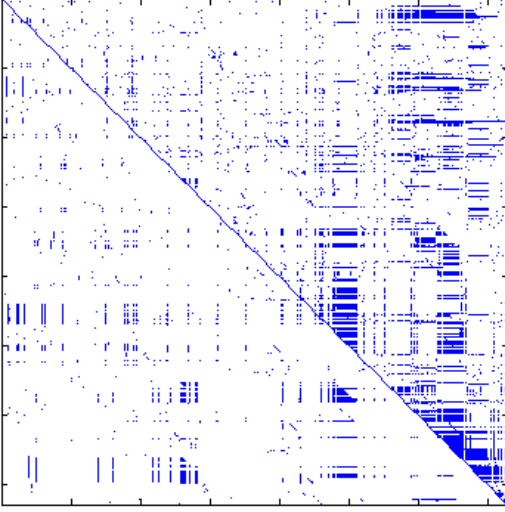
For example: Applying Gaussian elimination for $B = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 3 & 0 \\ -1 & 3 & 2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}$, the successive augmented matrices with the corresponding matrices of multipliers are

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 1 & 3 & 0 & 2 \\ -1 & 3 & 2 & 0 \end{array} \right] & L = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix} \\ \left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ & 2 & 1 & 0 \\ & 4 & 1 & 2 \end{array} \right] & L = \begin{bmatrix} 1 & & \\ & 1 & 1 \\ & -1 & 1 \end{bmatrix} \\ \left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ & 2 & 1 & 0 \\ & & -1 & 2 \end{array} \right] & L = \begin{bmatrix} 1 & & \\ & 1 & 1 \\ & -1 & 2 & 1 \end{bmatrix}, \end{aligned}$$

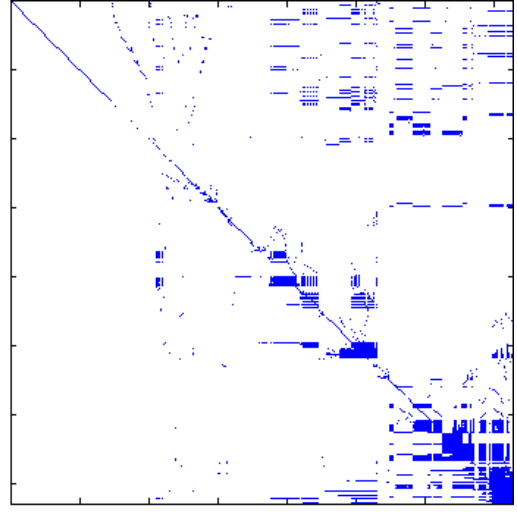
where some zero matrix entries are left blank to stress the structure. The solution of $B\mathbf{x} = \mathbf{b}$ is then found by solving $U\mathbf{x} = \mathbf{y}$ as $x_3 = 2/(-1) = -2$; $x_2 = (0 - (-2))/2 = 1$; $x_1 = (2 - 1 - (-1)(-2))/1 = -1$. It is readily verified that $B = LU$ for

$$L = \begin{bmatrix} 1 & & \\ 1 & 1 & \\ -1 & 2 & 1 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 1 & 1 & -1 \\ & 2 & 1 \\ & & -1 \end{bmatrix}$$

Given $B = LU$, the solution of any system $B\mathbf{x} = \mathbf{b}$ can be found by solving $L\mathbf{y} = \mathbf{b}$ then $U\mathbf{x} = \mathbf{y}$. That $\mathbf{x}$ thus formed is the solution of $B\mathbf{x} = \mathbf{b}$ follows by substituting $\mathbf{y} = U\mathbf{x}$ into $L\mathbf{y} = \mathbf{b}$, to give $LU\mathbf{x} = \mathbf{b}$ iff $B\mathbf{x} = \mathbf{b}$. The LU decomposition can be used to solve any system $B^T\mathbf{x} = \mathbf{b}$. The LU decomposition of a matrix B when not exploiting sparsity requires the calculation of m^2 values: $m(m-1)/2$ in L (since the unit diagonal entries are known) and $(m+1)m/2$ in U : no fewer than the number of entries in B^{-1} . However, when exploiting sparsity, the total number of nonzeros in L and U is expected to be very much less than the number of nonzeros in B^{-1} . This is because column j of U is the solution of $Lu_j = b_j$, where b_j is column j of B , whereas column j of B^{-1} is the solution of $Bx_j = e_j$. In the case of the basis matrix for STAIR in Figure 2a, the sparsity pattern of the decomposition $B = LU$ is illustrated in Figure 3a, where the sparsity patterns below the diagonal in L , and above the diagonal



(a) Standard LU decomposition has 7850 nonzeros, a density of 6.0% and fill factor of 2.4



(b) Markowitz LU decomposition has 5043 nonzeros, density 3.8% and fill factor of 1.5

Figure 3: Forming the LU decomposition of B for STAIR

in U are shown together as a single square matrix. Note that the fill factor is more than ten times less than the fill factor of B^{-1} in Figure 2b. The effect of using the “Markowitz” technique for finding the LU decomposition of a sparse matrix is illustrated in Figure 3b. The fill factor of 1.5 is two-thirds of the fill factor for the standard LU decomposition, and a fill factor of 1.0 is as good as can be expected. These fill factors are a measure of the time taken to compute a particular invertible representation of B , and to use it to solve systems of equations.

6.2 Updating the invertible representation of B

The cost of forming an LU decomposition of B is such that, when it is updated - by column p being replaced by the vector $\mathbf{a}_q$ - it is very important to avoid the cost of decomposing the new basis matrix. This is achieved by writing it as

$$B + (\mathbf{a}_q - B\mathbf{e}_p)\mathbf{e}_p^T = B[I + (\hat{\mathbf{a}}_q - \mathbf{e}_p)\mathbf{e}_p^T] = BE$$

where $E = I + (\hat{\mathbf{a}}_q - \mathbf{e}_p)\mathbf{e}_p^T$. The expression for E^{-1} obtained via the Sherman-Morrison formula allows operations using E^{-1} to be performed at a cost proportional to m (at worst) and only requires the vector $\hat{\mathbf{a}}_q$ and the index p to be stored. Hence operations with the inverse of the updated basis matrix can be performed using the invertible representations of B and E which, together are an invertible representation of the updated basis matrix. It follows, inductively, that operations with the inverse of all further updated basis matrices can be performed without reforming an LU decomposition. However, for reasons beyond the scope of this course, it is eventually preferable to form the LU decomposition of the basis matrix than continue updating indefinitely.

7 Sensitivity

The data used in an LP problem are not necessarily known precisely, or the modeller may want to know how the optimal solution and objective function value behave if certain values are changed. To address these challenges, the sensitivity of the solution to data is analysed.

7.1 RHS sensitivity

Although the right-hand-side (RHS) sensitivity analysis below applies to any LP problem, there is an important interpretation in the context of resource optimization problems,

$$\text{maximize } f = \mathbf{c}^T \mathbf{x} \quad \text{subject to } A\mathbf{x} \leq \mathbf{b} \quad \mathbf{x} \geq \mathbf{0}$$

where the RHS vector $\mathbf{b}$ contains the availability of resources. After adding slack variables, the LP in standard form is

$$\begin{aligned} \text{maximize } f &= \bar{\mathbf{c}}^T \mathbf{x} \\ \text{subject to } &\bar{A}\mathbf{x} = \mathbf{b} \\ &\mathbf{x} \geq \mathbf{0} \end{aligned}$$

Let $\mathcal{B}$ and $\mathcal{N}$ yield an optimal solution for which the value of the basic variables is $\hat{\mathbf{b}}$

Consider a change δ in the RHS for inequality i . This corresponds to a change δ in the availability of resource i . The corresponding RHS of the LP in standard form is $b_i + \delta$. As a whole, the LP itself has RHS $\mathbf{b} + \delta \mathbf{e}_i$, where $\mathbf{e}_i$ is column i of the identity matrix. It will be seen below that, if $|\delta|$ is sufficiently small (possibly zero) then $\mathcal{B}$ and $\mathcal{N}$ correspond to an optimal solution of the modified LP.

Substituting $\mathbf{b} + \delta \mathbf{e}_i$ for $\mathbf{b}$ in the expression for the value of the basic variables, The optimal values of the basic variables are given by

$$\mathbf{x}_B = B^{-1}(\mathbf{b} + \delta \mathbf{e}_i) = \hat{\mathbf{b}} + \delta B^{-1} \mathbf{e}_i$$

Since $\mathcal{N}$ does not change the nonbasic variables remain zero. This expression is valid so long as $\mathbf{x}_B$ is feasible, that is

$$\hat{\mathbf{b}} + \delta B^{-1} \mathbf{e}_i \geq \mathbf{0}$$

Observe that $B^{-1} \mathbf{e}_i$ is column i of B^{-1} . This yields an interval or **range** $\underline{\delta} \leq \delta \leq \bar{\delta}$ such that $\hat{\mathbf{b}} + \delta B^{-1} \mathbf{e}_i \geq \mathbf{0}$ where

$$\underline{\delta} = \max_{[B^{-1}]_{ji} > 0} -\frac{\hat{b}_j}{[B^{-1}]_{ji}} \quad \text{and} \quad \bar{\delta} = \min_{[B^{-1}]_{ji} < 0} -\frac{\hat{b}_j}{[B^{-1}]_{ji}}$$

Fair prices

Substituting $\mathbf{x}_B = \hat{\mathbf{b}} + \delta B^{-1} \mathbf{e}_i$ for $\mathbf{x}_B$ in the partitioned objective function gives

$$\begin{aligned} f &= \mathbf{c}_B^T (\hat{\mathbf{b}} + \delta B^{-1} \mathbf{e}_i) \\ &= \mathbf{c}_B^T \hat{\mathbf{b}} + \delta \mathbf{c}_B^T B^{-1} \mathbf{e}_i \\ &= \hat{f} + \delta \boldsymbol{\pi}^T \mathbf{e}_i \\ &= \hat{f} + \delta \pi_i \end{aligned}$$

So π_i is the increase in the objective for unit increase in the RHS of equation i . This is the **fair price** for more of the resource: if one more unit is available, the objective increases by π_i . It is the value of the resource.

- If the price of unit amount is more than π_i the net change in the objective is negative: buying more of the resource is unattractive.

- If the price of unit amount is less than π_i the net change in the objective is positive: buying more of the resource is attractive.

Fair price for basic slack

If the slack variable x_{n+i} for equation i is basic, suppose $n+i$ is entry j of $\mathcal{B}$ and consider equation j of $B^T \pi = c_B$

- Column j of B is e_i , so row j of B^T is e_i^T
- Entry j of c_B is zero since x_{n+i} is slack

Equation j of $B^T \pi = c_B$ is, thus, $e_i^T \pi = 0 \Rightarrow \pi_i = 0$, so the fair price for equation i is zero. In summary, if the slack is basic then $x_{n+i} > 0$ (unless $\hat{b}$ is degenerate), so not all of the resource in constraint i is used up. There is simply no fair price for more of the resource.

Fair price for nonbasic slack

If the slack variable x_{n+i} for equation i is nonbasic, suppose $n+i$ is entry j of $\mathcal{N}$ and consider its reduced cost $\hat{c}_j = [c_N]_j - [N]_j^T \pi$

- $[c_N]_j = 0$ since x_{n+i} is slack
- $[N]_j = e_i$ since x_{n+i} is the slack for equation i

Hence

$$\hat{c}_j = 0 - e_i^T \pi = -\pi_i \Rightarrow \pi_i = -\hat{c}_j \geq 0$$

since $\hat{c}_N \leq 0$.

Hence the fair price for equation i is the negated reduced cost of its slack variable x_{n+i} . In summary, if the slack is nonbasic then $x_{n+i} = 0$ so all of the resource in constraint i is used up, and the fair price for more of the resource is the (non-negative) reduced cost of its slack variable x_{n+i} .

Lower bound sensitivity

For a constraint with a lower bound of l_i , the RHS of the LP in standard form (after negating the inequality and adding a slack) is $-l_i$. Hence the sensitivity data relates to a RHS value of $-l_i + \delta$, thus the fair price corresponds to the lower bound being *reduced* by δ . Although this appears to be inconsistent, increases in upper bounds and decreases in lower bounds all increase the size of the feasible region, and in both cases the fair price is the value for this increase.

7.2 Cost sensitivity

If an LP cost coefficient is changed by a small amount, then the optimal solution is unaffected. However, eventually, a reduced cost will change sign, making the vertex non-optimal. If the cost is changed for a basic variable then one component of c_B changes so, via $\pi = B^{-T} c_B$, all the reduced costs change (in general). If the cost is changed for a nonbasic variable then c_B is unchanged, so only the reduced cost for that variable is changed.

7.3 Coefficient sensitivity

Matrix coefficient sensitivity is harder to analyse. If the coefficient for a nonbasic variable is changed then N changes, so one reduced cost will change. However, if the coefficient for a basic variable is changed then B changes. Not only does this change π and $\hat{b}$, but B may become singular. In the context of problems in two variables, the slope of a constraint line changes, rather than the intercepts, so the constraint line may become parallel to another constraint line.

Sensitivity example

Using the Scotfoods example from Workshop 1

Product	Proportion of each type of oil			Selling price (£/tonne)
	Olive	Corn	Sunflower	
Cooking oil	0.2	0.4	0.4	600
Salad oil	0.5	0.0	0.5	900
Supply (tonnes/day)	200	180	250	

- Decision variables
 - Let x_1 be the amount of cooking oil made (tonnes)
 - Let x_2 be the amount of salad oil made (tonnes)
- The LP to be solved (in standard form) is

$$\begin{aligned}
 \text{maximize } f &= 600x_1 + 900x_2 \\
 \text{subject to } &0.2x_1 + 0.5x_2 + x_3 = 200 \quad (\text{Oliveoil}) \\
 &0.4x_1 + x_4 = 180 \quad (\text{Cornoil}) \\
 &0.4x_1 + 0.5x_2 + x_5 = 250 \quad (\text{Sunfloweroil}) \\
 &x_1 \geq 0 \quad x_2 \geq 0 \quad x_3 \geq 0 \quad x_4 \geq 0 \quad x_5 \geq 0
 \end{aligned}$$

The optimal basis is $\mathcal{B} = \{1, 2, 4\}$ and $\mathcal{N} = \{3, 5\}$ so

$$B = \begin{bmatrix} 0.2 & 0.5 & 0 \\ 0.4 & 0 & 1 \\ 0.4 & 0.5 & 0 \end{bmatrix}$$

with optimal solution $\mathbf{x} = [250 \quad 300 \quad 0 \quad 80 \quad 0]^T$.

Consider increasing the RHS of equation 1 (increasing the availability of olive oil) so $\mathbf{b} = \begin{bmatrix} 200 + \delta \\ 180 \\ 250 \end{bmatrix}$

$$B^{-1}\mathbf{e}_1 = \begin{bmatrix} -5 \\ 4 \\ 2 \end{bmatrix} \quad \text{so solution is } \mathbf{x} = \begin{bmatrix} 250 \\ 300 \\ 0 \\ 80 \\ 0 \end{bmatrix} + \delta \begin{bmatrix} -5 \\ 4 \\ 0 \\ 2 \\ 0 \end{bmatrix}$$

for $\underline{\delta} \leq \delta \leq \bar{\delta}$, where

$$\bar{\delta} = \frac{-250}{-5} = 50 \quad \underline{\delta} = \max \left\{ \frac{-300}{4}, \frac{-80}{2} \right\} = -40$$

Substitute solution into the objective to give

$$f = 600(250 - 5\delta) + 900(300 + 4\delta) = 420000 + 600\delta$$

Fair price for more olive oil is £600/tonne.

In general

$$\mathbf{c}_B = \begin{bmatrix} 600 \\ 900 \\ 0 \end{bmatrix} \quad \text{so } \boldsymbol{\pi} = B^{-T}\mathbf{c}_B = \begin{bmatrix} 600 \\ 0 \\ 1200 \end{bmatrix}$$

- Nonbasic slack for olive oil so fair price ($\pi_1 = 600$) is £600/tonne.
- Basic slack for corn oil ($x_4 = 80 > 0$) so no fair price (and $\pi_2 = 0$).
- Nonbasic slack for sunflower oil so fair price is ($\pi_3 = 1200$) is £1200/tonne.

8 Duality

Duality is great! It's one of the few aspects of optimization that pure mathematicians find cool, but also has some properties that are of huge practical value.

8.1 Definition

For the following LP problem

$$\text{maximize } f = \mathbf{c}^T \mathbf{x} \quad \text{subject to } A\mathbf{x} \leq \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0} \quad (P)$$

The **dual** LP problem is

$$\text{minimize } f = \mathbf{b}^T \mathbf{y} \quad \text{subject to } A^T \mathbf{y} \geq \mathbf{c}, \quad \mathbf{y} \geq \mathbf{0} \quad (D)$$

Problem (P) is known as the **primal** problem. The two problems share the same data A , $\mathbf{b}$ and $\mathbf{c}$ but the primal problem has n variables and m constraints, and the dual problem has m variables and n constraints.

The dual of the dual is the primal

The dual LP problem

$$\text{minimize } f = \mathbf{b}^T \mathbf{y} \quad \text{subject to } A^T \mathbf{y} \geq \mathbf{c}, \quad \mathbf{y} \geq \mathbf{0} \quad (D)$$

is, equivalently

$$\text{maximize } f = -\mathbf{b}^T \mathbf{y} \quad \text{subject to } -A^T \mathbf{y} \leq -\mathbf{c}, \quad \mathbf{y} \geq \mathbf{0}$$

This is in primal form with constraint matrix $-A^T$; cost vector $-\mathbf{b}$; RHS vector $-\mathbf{c}$. Hence the dual of (D) is

$$\text{minimize } f = (-\mathbf{c})^T \mathbf{z} \quad \text{subject to } (-A^T)^T \mathbf{z} \geq (-\mathbf{b}), \quad \mathbf{z} \geq \mathbf{0}$$

which is, equivalently,

$$\text{maximize } f = \mathbf{c}^T \mathbf{z} \quad \text{subject to } A\mathbf{z} \leq \mathbf{b}, \quad \mathbf{z} \geq \mathbf{0} \quad (P)$$

This is the original primal problem.

Duality: Example 1

The ScotFoods model is

$$\begin{aligned} &\text{maximize} && 600x_1 + 900x_2 \\ &\text{subject to} && 0.2x_1 + 0.5x_2 \leq 200 \\ &&& 0.4x_1 \leq 180 \\ &&& 0.4x_1 + 0.5x_2 \leq 250 \\ &&& x_1 \geq 0 \quad x_2 \geq 0 \end{aligned} \quad (P)$$

The dual problem is

$$\begin{aligned} &\text{minimize} && 200y_1 + 180y_2 + 250y_3 \\ &\text{subject to} && 0.2y_1 + 0.4y_2 + 0.4y_3 \geq 600 \\ &&& 0.5y_1 + 0.5y_3 \geq 900 \\ &&& y_1 \geq 0 \quad y_2 \geq 0 \quad y_3 \geq 0 \end{aligned} \quad (D)$$

which is, equivalently,

$$\begin{aligned} &\text{maximize} && -200y_1 - 180y_2 - 250y_3 \\ &\text{subject to} && -0.2y_1 - 0.4y_2 - 0.4y_3 \leq -600 \\ &&& -0.5y_1 - 0.5y_3 \leq -900 \\ &&& y_1 \geq 0 \quad y_2 \geq 0 \quad y_3 \geq 0 \end{aligned}$$

with dual problem

$$\begin{aligned}
& \text{minimize} && -600z_1 & - & 900z_2 \\
& \text{subject to} && -0.2z_1 & - & 0.5z_2 & \geq & -200 \\
& && -0.4z_1 & & & \geq & -180 \\
& && -0.4z_1 & - & 0.5z_2 & \geq & -250 \\
& && z_1 \geq 0 & & z_2 \geq 0
\end{aligned}$$

which is, equivalently,

$$\begin{aligned}
& \text{maximize} && 600z_1 & + & 900z_2 \\
& \text{subject to} && 0.2z_1 & + & 0.5z_2 & \leq & 200 \\
& && 0.4z_1 & & & \leq & 180 \\
& && 0.4z_1 & + & 0.5z_2 & \leq & 250 \\
& && z_1 \geq 0 & & z_2 \geq 0
\end{aligned}$$

which is (P) , illustrating the property that *The dual of the dual is the primal*.

8.2 Weak duality

Theorem 6: Weak duality If $\mathbf{x}$ and $\mathbf{y}$ are feasible solutions of the primal and dual problems respectively then

$$\mathbf{c}^T \mathbf{x} \leq \mathbf{b}^T \mathbf{y}$$

Proof

$$\begin{aligned}
\mathbf{c}^T \mathbf{x} &= \mathbf{x}^T \mathbf{c} \\
&\leq \mathbf{x}^T (A^T \mathbf{y}) \quad [\text{Since } \mathbf{c} \leq A^T \mathbf{y} \Leftarrow A^T \mathbf{y} \geq \mathbf{c} \Leftarrow \mathbf{y} \text{ feasible}] \\
&= (\mathbf{x}^T A^T) \mathbf{y} \\
&= (A\mathbf{x})^T \mathbf{y} \\
&\leq \mathbf{b}^T \mathbf{y} \quad [\text{Since } A\mathbf{x} \leq \mathbf{b} \Leftarrow \mathbf{x} \text{ feasible}]
\end{aligned}$$

Consequences

Since $\mathbf{c}^T \mathbf{x} \leq \mathbf{b}^T \mathbf{y}$, if the primal problem (P) is unbounded then the dual problem (D) is infeasible. This follows by contradiction. Let $\mathbf{y}$ be a feasible solution of (D) , then $\mathbf{b}^T \mathbf{y}$ is an upper bound on the primal objective, contradicting its unboundedness.

Similarly, if the dual problem (D) is unbounded then the primal problem (P) is infeasible

Duality: Example 2

The following example of a primal problem (P) and its dual (D) illustrates the property that (P) unbounded $\Rightarrow (D)$ infeasible which follows as a consequence of the weak duality theorem [Theorem 6] that if $\mathbf{x}$ and $\mathbf{y}$ are feasible solutions of the primal and dual problems respectively then $\mathbf{c}^T \mathbf{x} \leq \mathbf{b}^T \mathbf{y}$. For the following LP

$$\text{maximize } f = \begin{bmatrix} 1 \\ 2 \end{bmatrix}^T \mathbf{x} \quad \text{subject to } \begin{bmatrix} -1 & -1 \end{bmatrix} \mathbf{x} \leq \begin{bmatrix} -1 \end{bmatrix} \quad \mathbf{x} \geq \mathbf{0} \quad (P)$$

$A = \begin{bmatrix} -1 & -1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} -1 \end{bmatrix}$ and $\mathbf{c} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ so the corresponding dual LP problem is

$$\text{minimize } f = -y \quad \text{subject to } \begin{bmatrix} -1 \\ -1 \end{bmatrix} y \geq \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad y \geq 0 \quad (D)$$

or, equivalently

$$\text{maximize } f = y \quad \text{subject to } \begin{bmatrix} 1 \\ 1 \end{bmatrix} y \leq \begin{bmatrix} -1 \\ -2 \end{bmatrix} \quad y \geq 0$$

Observe that

- (P) is unbounded [since arbitrarily large values of x_1 and x_2 are feasible and the objective has positive coefficients]
- (D) is infeasible [since y cannot satisfy both $y \geq 0$ and $y \leq -2$]

Further, note that the dual of (D) is

$$\text{minimize } f = \begin{bmatrix} -1 \\ -2 \end{bmatrix}^T \mathbf{z} \quad \text{subject to } \begin{bmatrix} 1 & 1 \end{bmatrix} \mathbf{z} \geq \begin{bmatrix} 1 \end{bmatrix} \quad \mathbf{z} \geq \mathbf{0}$$

or, equivalently

$$\text{maximize } f = \begin{bmatrix} 1 \\ 2 \end{bmatrix}^T \mathbf{z} \quad \text{subject to } \begin{bmatrix} -1 & -1 \end{bmatrix} \mathbf{z} \leq \begin{bmatrix} -1 \end{bmatrix} \quad \mathbf{z} \geq \mathbf{0}$$

which is (P) , illustrating the property that *The dual of the dual is the primal*.

8.3 Strong duality

As the name suggests, strong duality links the primal and dual LPs in a more fundamental way than weak duality. As well as being mathematically elegant, it is of huge practical value. The mathematical result is established in Theorem 7. The mathematical essence of strong duality is that the $\boldsymbol{\pi}$ vector for an optimal solution of the primal problem is an optimal solution of the dual problem - and vice versa. The proof is nontrivial, but given for completeness since this is such a valuable result. The strong duality result means that only one of (P) or (D) needs to be solved, and is one of many consequences that are of practical value.

Theorem 7: Strong duality

For the primal LP

$$\text{maximize } f = \mathbf{c}^T \mathbf{x} \quad \text{subject to } A\mathbf{x} \leq \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0} \quad (P)$$

If $\mathbf{x}^*$ is an optimal (basic) solution for the problem in standard form then the optimal solution of

$$\text{minimize } f = \mathbf{b}^T \mathbf{y} \quad \text{subject to } A^T \mathbf{y} \geq \mathbf{c}, \quad \mathbf{y} \geq \mathbf{0} \quad (D)$$

is $\mathbf{y}^* = \boldsymbol{\pi} = B^{-T} \mathbf{c}_B$, and the optimal objective values are equal: $\mathbf{c}^T \mathbf{x}^* = \mathbf{b}^T \mathbf{y}^*$

Proof:

For convenience, suppose that slacks $\mathbf{s}$ are added to the inequalities in (P) and slacks $\mathbf{z}$ are added to the inequalities in (D) to put them in standard form

$$\text{maximize } \mathbf{c}^T \mathbf{x} \quad \text{subject to } A\mathbf{x} + \mathbf{s} = \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0} \quad \mathbf{s} \geq \mathbf{0} \quad (P)$$

$$\text{minimize } \mathbf{b}^T \mathbf{y} \quad \text{subject to } A^T \mathbf{y} - \mathbf{z} = \mathbf{c}, \quad \mathbf{y} \geq \mathbf{0} \quad \mathbf{z} \geq \mathbf{0} \quad (D)$$

Now

$$\mathbf{c}^T \mathbf{x} = \mathbf{x}^T \mathbf{c} = \mathbf{x}^T (A^T \mathbf{y} - \mathbf{z}) = (A\mathbf{x})^T \mathbf{y} - \mathbf{x}^T \mathbf{z} = (\mathbf{b} - \mathbf{s})^T \mathbf{y} - \mathbf{z}^T \mathbf{x} = \mathbf{b}^T \mathbf{y} - \mathbf{s}^T \mathbf{y} - \mathbf{z}^T \mathbf{x}$$

So, since all of $\mathbf{x}$, $\mathbf{s}$, $\mathbf{y}$ and $\mathbf{z}$ are non-negative, $\mathbf{s}^T \mathbf{y} \geq 0$ and $\mathbf{z}^T \mathbf{x} \geq 0$ so $\mathbf{c}^T \mathbf{x} \leq \mathbf{b}^T \mathbf{y}$ (Weak duality).

The first mathematical aim is to show that, for an optimal solution of (P) , a vector $\mathbf{y}$ can be constructed so that $\mathbf{s}^T \mathbf{y} = \mathbf{z}^T \mathbf{x} = 0$, hence $\mathbf{c}^T \mathbf{x} = \mathbf{b}^T \mathbf{y}$.

Suppose that $\mathbf{x}^*$ is an optimal basic solution for (P) with reduced costs $\widehat{\mathbf{c}}_N \leq \mathbf{0}$ satisfying $N^T \boldsymbol{\pi} = \mathbf{c}_N - \widehat{\mathbf{c}}_N$

Claim: $\boldsymbol{\pi}$ is feasible for (D)

Firstly, $\boldsymbol{\pi} \geq \mathbf{0}$ (since $\boldsymbol{\pi}$ are fair prices, so non-negative), so $\mathbf{y} = \boldsymbol{\pi}$ satisfies the bounds of (D) .

Secondly, since $B^T \boldsymbol{\pi} = \mathbf{c}_B$ and $N^T \boldsymbol{\pi} = \mathbf{c}_N - \widehat{\mathbf{c}}_N$,

$$\begin{bmatrix} B^T \\ N^T \end{bmatrix} \boldsymbol{\pi} = \begin{bmatrix} \mathbf{c}_B \\ \mathbf{c}_N \end{bmatrix} - \begin{bmatrix} \mathbf{0} \\ \widehat{\mathbf{c}}_N \end{bmatrix}$$

Hence, considering the rows corresponding to original variables, $A^T \boldsymbol{\pi} = \mathbf{c} + \mathbf{z}$ ($\Rightarrow A^T \boldsymbol{\pi} - \mathbf{z} = \mathbf{c}$) where, since $\widehat{\mathbf{c}}_N \leq \mathbf{0}$, components of $\mathbf{z}$ are $z_i = 0$ for $i \in \mathcal{B}$ and $z_i = -\widehat{c}_i \geq 0$ for $i \in \mathcal{N}$, so $\mathbf{z} \geq \mathbf{0}$ (and $A^T \boldsymbol{\pi} \geq \mathbf{c}$). So $\boldsymbol{\pi}$ with $\mathbf{z} \geq \mathbf{0}$ satisfies the equations of (D)

Hence $\boldsymbol{\pi}$ is feasible for (D)

Now for an optimal solution $\mathbf{x}^*$ of (P) consider values of slack s_i for $i = 1, \dots, m$. Firstly, $s_i \geq 0$ since $\mathbf{x}^*$ is feasible

If $s_i > 0$ then it is a basic variable, so $\pi_i = 0$ (Fair price of basic slack is zero)

Otherwise $s_i = 0$

Hence $\mathbf{s}^T \mathbf{y} = 0$

For $\mathbf{x}^*$ itself,

If $x_i > 0$ then it is a basic variable ($i \in \mathcal{B}$), so $z_i = 0$ (as shown above)

Otherwise $x_i = 0$

Hence $\mathbf{z}^T \mathbf{x} = 0$ and, since $\mathbf{s}^T \mathbf{y} = 0$, $\mathbf{c}^T \mathbf{x} = \mathbf{b}^T \mathbf{y}$.

The consequence of $\mathbf{b}^T \mathbf{y} = \mathbf{c}^T \mathbf{x}^*$ for $\mathbf{y} = \boldsymbol{\pi}$ is that there is no feasible value of $\mathbf{y}$ with greater objective value. Hence $\mathbf{y}^* = \boldsymbol{\pi}$ solves (D) .

Conclusions

Since the optimal solution (D) may be deduced from the optimal solution of (P) , the property that the dual of (D) is (P) implies that the optimal solution $\mathbf{x}^*$ of (P) may be deduced from the optimal solution of (D) . Hence only one of (P) and (D) need ever be solved.

Theorem 7 leads to the (practically preferred) **dual simplex algorithm**, which is the (primal) simplex algorithm applied to (D) . The dual simplex algorithm operates in space of dual variables. Implementations of the dual simplex algorithm are usually faster than primal simplex implementations, partly because it is usually easier to find a dual feasible solution of practical LP problems than a primal feasible solution.

Example

For the ScotFoods model, the optimal primal solution is $\mathbf{x}^* = \begin{bmatrix} 250 \\ 300 \end{bmatrix}$ so $\mathbf{c}^T \mathbf{x}^* = 420000$ with

$$\mathbf{s} = \begin{bmatrix} 0 \\ 80 \\ 0 \end{bmatrix}, \quad \boldsymbol{\pi} = \begin{bmatrix} 600 \\ 0 \\ 1200 \end{bmatrix} \quad \text{and} \quad \widehat{\mathbf{c}}_N = \begin{bmatrix} -600 \\ -1200 \end{bmatrix}$$

For $\mathbf{y}^* = \boldsymbol{\pi}$, $\mathbf{z} = A^T \boldsymbol{\pi} - \mathbf{c} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ so $\mathbf{s}^T \mathbf{y}^* = 0$, $\mathbf{z}^T \mathbf{x}^* = 0$ and $\mathbf{b}^T \mathbf{y}^* = 420000 = \mathbf{c}^T \mathbf{x}^*$.

8.4 Application: Adding a constraint to the primal problem

Suppose that the primal LP

$$\text{maximize } f = \mathbf{c}^T \mathbf{x} \quad \text{subject to } A\mathbf{x} \leq \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0} \quad (P)$$

has been solved. Consider adding the constraint $\mathbf{a}^T \mathbf{x} \leq b$ to form the problem

$$\text{maximize } f = \mathbf{c}^T \mathbf{x} \quad \text{subject to } \begin{bmatrix} A \\ \mathbf{a}^T \end{bmatrix} \mathbf{x} \leq \begin{bmatrix} \mathbf{b} \\ b \end{bmatrix}, \quad \mathbf{x} \geq \mathbf{0} \quad (P')$$

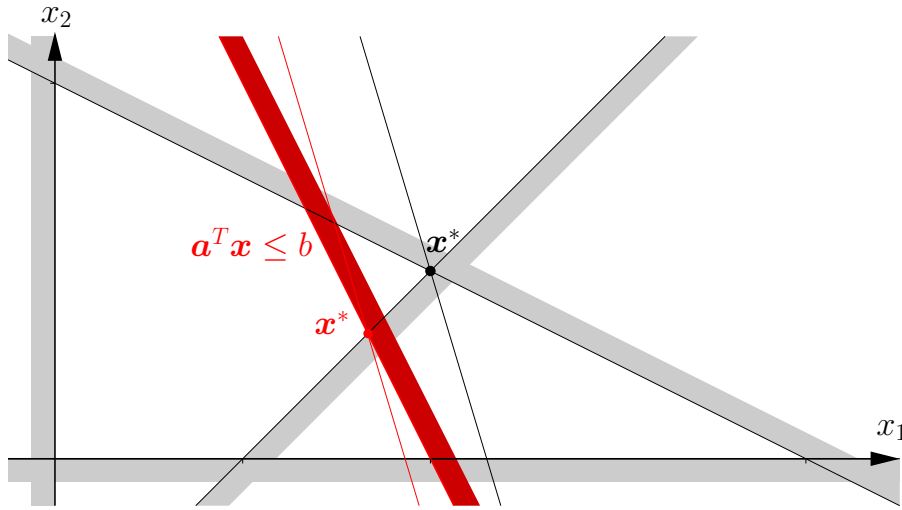


Figure 4: Cutting off the optimal solution when adding constraint $\mathbf{a}^T \mathbf{x} \leq b$ to a solved LP

This is a **tightening** of (P) , so called because (P') has the same variables but more constraints than (P) . An optimal solution $\mathbf{x}^*$ of (P) either

- Satisfies $\mathbf{a}^T \mathbf{x}^* \leq b$, so is a feasible solution of (P') , so $\mathbf{x}^*$ solves (P')
- Does not satisfy $\mathbf{a}^T \mathbf{x}^* \leq b$, so is not a feasible solution of (P') , as illustrated in Figure 4

So feasibility must be restored to solve (P')

The dual problem corresponding to (P') is

$$\text{minimize } f = \mathbf{b}^T \mathbf{y} + by \quad \text{subject to } A^T \mathbf{y} + \mathbf{a}y \geq \mathbf{c}, \quad \mathbf{y} \geq \mathbf{0} \quad y \geq 0 \quad (D')$$

This is a **relaxation** of (D) , so called because (D') has more variables but the same number of constraints as (D) . Given an optimal solution $\mathbf{y}^*$ for (D)

- y^* and $y = 0$ are a feasible solution of (D')
Only have to find an optimal solution of (D')

Hence, after tightening the primal problem, it's easier to solve (D') and deduce the solution to (P') than it is to solve (P') directly. Solving after adding one/more constraints is important in **branch-and-bound** for integer programming. Retaining dual feasibility makes it more attractive to solve the dual problem or use the dual simplex method.